

Three claims of unequal strength: partial deafferentation, central hyperexcitability and body disownership in hard flaccid syndrome

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Twelfth in a series; see companion papers on the neural, effector, sensory and structural conjectures.

Abstract

Background. The sensory picture of hard flaccid syndrome (HFS) contains an apparent contradiction. Numbness or loss of sensation was reported by 74.8% of surveyed patients — a *negative* sensory sign — while expert consensus places HFS within the persistent genital arousal disorder / genito-pelvic dysesthesia spectrum, characterised by sensory *hyperactivity*. The conjecture examined here resolves this by proposing partial deafferentation: one lesion producing reduced input and increased central gain simultaneously.

The resolution is sound, and is the conjecture's real contribution. Coexisting numbness and dysesthesia in the same territory is the ordinary signature of a partial nerve lesion, not a paradox requiring two mechanisms. This point is worth more than the rest of the conjecture combined and generalises beyond HFS.

But the conjecture bundles three claims of very unequal strength. Claim A — partial axonal loss causes the numbness — is well supported in principle and testable by territorial mapping. Claim B — central hyperexcitability in the partially deafferented territory causes the dysesthesia — is plausible but duplicates two companion hypotheses. Claim C — degraded cortical representation causes the reported feeling that the penis is hollow, disconnected or not part of the body — rests on the maladaptive cortical plasticity model, which has been under serious and unresolved challenge since 2013: phantom pain has been shown to be associated with *preserved* rather than remapped cortical representation.

Two things the conjecture does not consider, both cheaply checkable. Depersonalisation–derealisation disorder is a defined condition whose phenomenology matches the body-disownership report almost exactly — detachment from one's body, somatosensory distortion, reduced pain sensitivity — with typical onset at 16–23 years, stress precipitation, and a validated questionnaire. Seventy-five per cent of the largest HFS series had a history of depression or anxiety. Separately, 45.5% of that series were *actively using* anxiolytics or antidepressants, and persistent genital hypoesthesia is the cardinal symptom of post-SSRI sexual dysfunction, for which diagnostic criteria exist. Neither possibility is excluded by any published HFS study.

Conclusion. Claim A should be tested and probably retained. Claim C should be reframed as a differential diagnosis rather than a mechanism, and the two instruments needed — a sensory map and a questionnaire — are among the cheapest in this series.

Keywords: hard flaccid syndrome; deafferentation; genital numbness; cortical reorganisation; depersonalisation–derealisation; post-SSRI sexual dysfunction; quantitative sensory testing.

Reader advisory

This is hypothesis generation, not a report of new data. This paper raises two conditions — depersonalisation–derealisation disorder and post-SSRI sexual dysfunction — as possible contributors to or differentials for symptoms reported in hard flaccid syndrome. Neither observation is a diagnosis of anyone, and neither implies that reported symptoms are imagined; both are real conditions with real mechanisms. **Nobody should start, stop or change an antidepressant on the basis of this article.** Nothing here should influence clinical management.

1. Introduction

The sensory features of hard flaccid syndrome (HFS) are among its most consistently reported and least investigated. Numbness or loss of sensation somewhere on the penis was reported by 74.8% of 143 surveyed patients [B] [1]; pain, dysesthesia and paresthesia are described across the case literature [2]; and patients describe the penis as feeling hollow, disconnected, or not part of the body [E] [6].

There is a tension in how this is classified. Expert consensus places HFS within the persistent genital arousal disorder

/ genito-pelvic dysesthesia spectrum **[B]** [4, 5], a grouping characterised by sensory hyperactivity. Yet the dominant sensory complaint here is sensory *loss*.

The conjecture examined here resolves that tension elegantly: a partial lesion of the dorsal nerve of the penis produces axonal loss — hence numbness — while the partially deafferented central neurons become hyperexcitable, generating spontaneous and evoked abnormal sensation in the same territory. Reduced input and increased gain are two consequences of one lesion [6].

We think that resolution is correct and is the conjecture’s real contribution. We also think the conjecture bundles it with two further claims of much lower strength, and that the weakest of the three — the account of body disownership — rests on a model that has been contested for over a decade and competes with a defined psychiatric condition that nobody has looked for.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *deafferentation pain cortical reorganisation, maladaptive plasticity phantom limb, depersonalisation derealisation body detachment, post-SSRI sexual dysfunction genital numbness, quantitative sensory testing genital*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for the current status of the cortical remapping model and for alternative explanations of body-ownership disturbance, and Sections 4.1 and 5 report what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked **[VERIFY]**; quantities that do not exist in the literature are marked **[DATA NEEDED]**.

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, or a clinical entity defined in current diagnostic classifications.
[B]	Single human study, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. The part that is right

A partial lesion divides the afferent population. Destroyed axons produce negative signs — numbness, and loss of temperature, pressure, vibration and texture discrimination. Surviving injured axons and their uninjured neighbours become hyperexcitable and produce positive signs — paresthesia, dysesthesia, pain **[D]** [6, 8].

Two features of this deserve emphasis. First, **the coexistence of numbness and abnormal sensation in the same territory is not a contradiction but the expected signature of a partial nerve lesion**, and this is a general point about sensory neurology that the field appears not to have applied here. Second, it dissolves the classification tension: a spectrum defined by sensory hyperactivity can accommodate a condition whose dominant complaint is sensory loss, because both arise from the same lesion **[D]**.

The dorsal nerve of the penis is the terminal sensory branch of the pudendal nerve and its peripheral distribution in humans has been mapped **[B]** **[VERIFY]** [9]; the glans is densely supplied by free nerve endings **[B]** **[VERIFY]** [10]. That distribution is what makes Claim A testable: a partial lesion should produce a deficit that *respects* the nerve’s territory.

4. Three claims, not one

We separate the conjecture into its components because they do not stand or fall together, and because presenting them as a single hypothesis makes the whole appear as strong as its strongest part.

4.1 Claim C rests on a model that has been under challenge since 2013

The conjecture’s account of body disownership is the maladaptive plasticity model: a cortical representation deprived of reliable input degrades, neighbouring representations encroach, and the organ’s place in the body schema becomes

Table 2. The three claims bundled in the conjecture, separated by strength.

	Claim	Strength	Assessment
A	Partial axonal loss in the dorsal penile nerve causes the numbness and modality-specific sensory loss	[D], well grounded	Testable now by modality-by-modality mapping against the nerve's distribution. Retain and test.
B	Central hyperexcitability in the partially deafferented territory causes dysesthesia and paresthesia	[D], plausible	Duplicates the central sensitization and ectopic generator conjectures [7, 8]; adds no discriminating prediction of its own.
C	Degraded cortical representation causes the feeling that the penis is hollow, disconnected or not part of the body	[D], weak	Rests on a contested model (Section 4.1) and competes with a defined clinical condition nobody has excluded (Section 5).

unstable [6]. That model dominated deafferentation research for two decades. It is now the subject of a substantial and unresolved dispute.

The original evidence was that remapping of the lower face representation into the deprived hand territory correlated with phantom limb pain in upper limb amputees [B] [VERIFY] [13, 14]. The challenge is that phantom pain has been shown to be associated with *preserved* rather than degraded structure and function in the former hand area, together with *reduced inter-regional functional connectivity* — and that phantom pain appeared independent of cortical remapping [B] [15, 16]. The authors called for a reassessment of maladaptive plasticity theories ascribing a causal role to cortical remapping in chronic pain [16].

The debate remains open, with evidence and methodological criticism on both sides [B] [VERIFY] [17, 18]. Two implications for the conjecture [D]:

1. **The inference from deafferentation to degraded representation to disturbed body schema cannot be treated as established**, and a hypothesis about a genital symptom should not rest on the contested step of a limb literature.
2. **One element of the challenge may nonetheless help.** Makin and colleagues report that loss of sensory input is generally characterised by structural and functional degeneration in the deprived sensorimotor cortex, and that pain is associated with reduced *inter-regional connectivity* [15]. A disconnection finding is arguably a better fit for a symptom of felt disconnection than a remapping finding is — but this is our speculation, has not been examined in genital representation, and we flag it as such.

5. Depersonalisation: the differential the conjecture does not raise

The conjecture describes body disownership as “the most speculative claim in the article”, included because the symptom is reported consistently and nothing else accounts for it [6]. Something else does.

Depersonalisation–derealisation disorder is a dissociative disorder in current diagnostic classifications, consisting of persistent or recurrent feelings of being detached from one’s body or mental processes [A] [VERIFY] [19]. Its recognised symptom clusters include a sense of unreality about oneself *including detachment from one’s body*; affective blunting and *reduced pain sensitivity*; and perceptual disturbances including *tactile and somatosensory distortions* [B] [VERIFY] [20].

The epidemiological fit is uncomfortably close [D]:

- **Age.** Typical onset is between 16 and 23 years [B] [VERIFY] [21]; the mean age in the largest HFS survey was 27.4 years [1], and the reported age range across studies begins at 16 [2].
- **Precipitation.** Episodes are triggered by stress, depression, anxiety, or use of certain drugs [A] [VERIFY] [19]. Depression or anxiety was reported in 75% of the largest HFS series [B] [3].
- **Prevalence.** Between 25% and 75% of the general population have had at least one transient depersonalisation or derealisation experience; approximately 1–2% meet criteria for the disorder [A] [VERIFY] [19]. **The base rate of the transient experience is high enough that its appearance in a distressed, self-scrutinising population requires no special mechanism.**
- **Course.** Chronic depersonalisation is typically continuous with little fluctuation and independent of affective symptoms [B] [VERIFY] [21] — which matches the persistent, non-fluctuating quality patients describe.

There is also a counter-prediction worth recording. Depersonalisation disorder has been characterised as involving *reduced* psychogenic sympathoexcitation and blunted sympathetic responses [B] [VERIFY] [21]. If depersonalisation contributes materially to the HFS phenotype, that sits awkwardly with the sympathetic overactivity premise shared by most of this series [D].

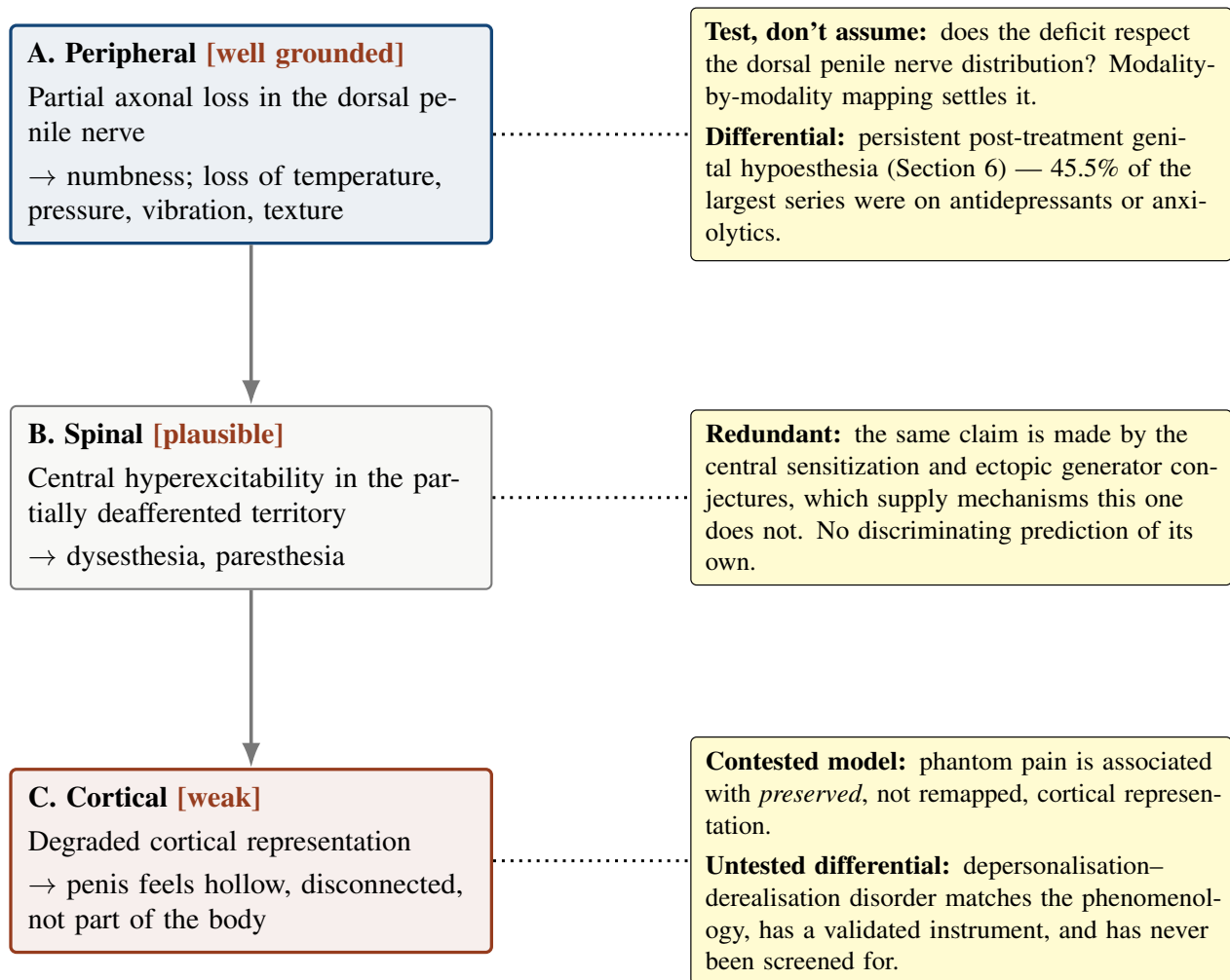


Figure 1. The conjecture's three claims, separated by strength, with what each needs. Presenting them as one hypothesis makes the whole appear as strong as Claim A. Claim A is well grounded and directly testable. Claim B duplicates two companion conjectures without adding a discriminating prediction. Claim C rests on a model under active challenge and has an untested clinical differential that matches its phenomenology closely.

5.1 Why this matters and what it does not mean

We want to be precise, because this territory is easily misread.

Raising depersonalisation **does not** imply that HFS is psychological, that the sensory symptoms are imagined, or that patients are misreporting. Depersonalisation is a real condition with real neurobiology, it is commonly precipitated by physical illness and distress, and its presence would be a *consequence* of a distressing genital condition at least as plausibly as a cause of one.

What it means is narrower and entirely methodological: **a symptom that is fully accounted for by a defined, common, instrument-measurable condition should not be used as evidence for a speculative cortical mechanism until that condition has been excluded.** The Cambridge Depersonalisation Scale is a validated instrument [B] [VERIFY] [22] and takes minutes to administer. No HFS study has reported it.

6. Post-SSRI sexual dysfunction: the differential nobody has excluded

A second, wholly separate possibility follows from a number in the same series.

In the largest reported HFS cohort, 75% had a history of depression or anxiety and **45.5% were actively using anxiolytics or antidepressants** [B] [3, 2]. Two of the two patients in one published case report received psychotropic medication [2].

Post-SSRI sexual dysfunction is an iatrogenic syndrome following cessation of serotonergic antidepressants, whose symptoms include **genital anaesthesia**, erectile dysfunction and orgasmic anhedonia, with additional non-sexual features including anhedonia, blunted affect and *depersonalisation* [B] [VERIFY] [23]. Persistent post-treatment genital

hypoesthesia has been surveyed as a distinct entity in past antidepressant users [B] [VERIFY] [24], and diagnostic criteria have been published [B] [VERIFY] [23].

The overlap with the HFS sensory picture is substantial: genital numbness, erectile difficulty, reduced erogenous sensation, blunted affect and a feeling of bodily detachment, in young men, many of whom have taken or are taking serotonergic antidepressants. **No published HFS study reports medication history in sufficient detail to exclude it,** and the diagnostic criteria have never been applied in this population [DATA NEEDED].

Again the point is methodological rather than dismissive. We are not proposing that HFS is post-SSRI sexual dysfunction — the hard flaccid state itself is not a recognised feature of it, and antidepressant exposure is common enough in any distressed young cohort to appear by chance. We are proposing that **a syndrome whose defining sensory symptom is genital numbness, in a cohort of whom nearly half are on serotonergic drugs, cannot be characterised without recording exposure and applying the existing criteria.**

7. The hypothesis

Formal statement, with the claims separated

H12-A (peripheral, recommended). Partial axonal loss in the dorsal nerve of the penis produces the negative sensory signs of hard flaccid syndrome. Prediction: modality-by-modality quantitative sensory testing shows a deficit that respects the anatomical distribution of that nerve.

H12-B (spinal, redundant). Central hyperexcitability in the partially deafferented territory produces the positive sensory signs. Prediction: extraterritorial spread of abnormality. Identical to the prediction of two companion conjectures [7, 8], which supply mechanisms this claim does not.

H12-C (cortical, not recommended in its current form). Degraded cortical representation produces the body disownership symptom. Prediction: altered genital representation on functional imaging. Rests on a contested model; competes with an unexcluded clinical differential.

H12-null. The sensory deficit does not respect the nerve territory, and the disownership symptom is accounted for by depersonalisation, medication exposure, or both.

Auxiliary assumptions:

- **A1 (a nerve lesion occurred).** Only 58.0% of surveyed patients reported onset after a specific incident [B] [1]. H12-A requires an injury in the remaining 42.0% that has not been identified.
- **A2 (the deficit is real).** Genital numbness in this population is established only by self-report; it has never been measured by quantitative sensory testing or by intraepidermal nerve fibre density [DATA NEEDED] [8].
- **A3 (the reports are not confounded).** That the sensory picture is not substantially produced by medication exposure or by depersonalisation. Currently unexamined.

8. Predictions and tests

8.1 The territorial mapping study

Prediction P1. In men with HFS, quantitative sensory testing performed modality by modality shows elevated detection thresholds for thermal, mechanical and vibratory modalities within the distribution of the dorsal nerve of the penis [9], with the boundary of the deficit corresponding to that distribution.

Prediction P2. Positive sensory phenomena — paresthesia, dysesthesia, secondary hyperalgesia — extend *beyond* that distribution.

The combination is what discriminates. **A deficit confined to the nerve territory with positive phenomena spreading beyond it is the signature of a partial peripheral lesion with central amplification,** which is exactly what the conjecture proposes. A deficit that ignores the territory indicates something else. A standardised, multicentre-validated testing battery with published reliability exists [A] [11, 12], though its reference values are for face, hand and foot and genital control data would have to be generated [DATA NEEDED].

8.2 The two questionnaire studies, which cost nothing

Prediction P3. Depersonalisation scores in men with HFS, measured with a validated instrument [22], exceed those of matched controls, and the body-disownership symptom is associated with the depersonalisation score rather than with the extent of sensory loss.

If disownership tracks a depersonalisation score and not the sensory map, Claim C is answered without any imaging.

If it tracks the sensory map and not the depersonalisation score, Claim C survives a real test. **This is the cheapest discriminating study proposed anywhere in this series.**

Prediction P4. Detailed serotonergic antidepressant exposure history, and application of published post-SSRI sexual dysfunction criteria, will identify a subgroup within HFS cohorts whose sensory symptoms are attributable to that exposure.

Both require only that investigators ask. Neither has been done.

8.3 What we do not recommend

Functional imaging of genital cortical representation would be the direct test of Claim C. We advise against it as a next step: the interpretive framework is contested [15, 16], genital representation in the human somatosensory cortex is less well characterised than hand representation [DATA NEEDED], the study would be expensive, and the two questionnaire studies above would render it either unnecessary or far better designed.

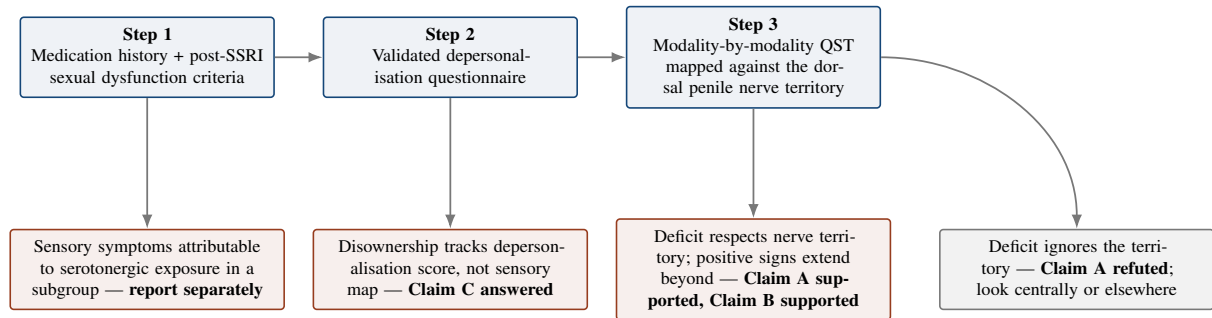


Figure 2. Proposed sequence for the sensory picture, ordered by cost. Steps 1 and 2 require only that investigators ask questions and administer a questionnaire, and both address confounds that no published study has excluded. Step 3 requires a validated sensory testing battery and locally generated genital control data. Functional imaging of cortical representation is deliberately absent: it is expensive, its interpretive framework is contested, and Step 2 would render it either unnecessary or much better designed.

Table 3. Predictions compared with the principal alternatives for the sensory picture.

Observation	H12 (deafferentation)	Depersonalisation	Serotonergic exposure	Discriminating value
Sensory deficit respects the dorsal penile nerve territory	Predicted	Not predicted	Not predicted	High
Positive phenomena extend beyond the territory	Predicted	Not predicted	Not predicted	Moderate
Disownership correlates with depersonalisation score	Not predicted	Predicted	Predicted	High
Disownership correlates with extent of sensory loss	Predicted	Not predicted	Not predicted	High
Genital numbness in men with no injury history	Not predicted	Possible	Predicted	Moderate
Blunted rather than elevated sympathetic reactivity	Not predicted	Predicted	Not specified	Moderate; cuts against the whole series
Symptoms in men with no antidepressant exposure	Predicted	Predicted	Refuted	High for that differential

9. Alternative explanations

1. **Depersonalisation–derealisation** (Section 5). Matches the disownership phenomenology, has a high base rate, is precipitated by stress and by distressing illness, and has never been screened for.
2. **Serotonergic antidepressant exposure** (Section 6). Genital hypoesthesia is its cardinal symptom; nearly half the largest cohort were taking such drugs.
3. **A peripheral ectopic generator without any central component** [8]. Produces both numbness and positive phenomena from partial axonal loss alone, and does not require Claims B or C.
4. **Hypervigilant attention.** Sustained scrutiny of genital sensation reliably produces altered perception of it. This is not a dismissal — attentional amplification is a real and measurable phenomenon — but it means self-reported sensory change in a closely monitored organ is weak evidence, and blinded threshold testing rather than symptom rating is the only way to separate them.

5. **The numbness may not be measurable.** Self-reported genital numbness has never been confirmed by quantitative testing in this population. It is possible that on formal examination thresholds are normal.

10. Implications

No clinical implications are drawn, and in particular nothing here is a reason for anyone to alter psychiatric medication. Three research implications follow.

First, unbundle the conjecture. Claim A deserves testing on its own terms. Claim B should be folded into the conjectures that supply its mechanism. Claim C should be reframed as a differential to be excluded rather than a mechanism to be defended.

Second, ask the two questions. Every HFS cohort study should record serotonergic antidepressant exposure in detail and administer a validated depersonalisation instrument. Both are free and both address confounds that currently make the sensory literature uninterpretable.

Third, map the territory. Whether the sensory deficit respects the distribution of the dorsal nerve of the penis is the single most informative sensory measurement available, and it discriminates across at least four papers in this series.

11. Limitations

1. **No evidence in HFS.** No quantitative sensory testing, no depersonalisation screening and no systematic medication history has been published in this population.
2. **The numbness is self-reported** and has never been objectively confirmed.
3. **Claim C rests on a contested model**, and our suggestion that a disconnection finding might fit better is speculation on our part with no supporting data.
4. **Genital cortical representation is poorly characterised** relative to limb representation, so even a well-designed imaging study would face interpretive difficulty.
5. **Raising two psychiatric and iatrogenic differentials risks misreading.** We have tried to state carefully that neither implies symptoms are imagined, but the distinction is easily lost and this is a population already poorly served by dismissal.
6. **The depersonalisation literature was accessed largely through reference texts and reviews**, and the specific figures quoted are flagged accordingly.
7. **The post-SSRI sexual dysfunction literature is young**, its criteria are recent, and its relationship to depression-related sexual dysfunction is contested.
8. **Claim A does not explain 42% of patients** who report no inciting incident.
9. **Several citations are flagged unverified**, including the phantom limb primary sources on both sides of the dispute.
10. **Confirmation risk.** This paper agrees with the conjecture's central insight and then argues that two thirds of it should be abandoned or reclassified. Readers should check the depersonalisation phenomenology against the reported HFS symptom themselves before accepting our claim that they match.

12. Refutation criteria

- **Claim A refuted.** Quantitative sensory testing shows a deficit that does not respect the distribution of the dorsal nerve of the penis, or shows no measurable deficit at all.
- **Claim A weakened.** Penile intraepidermal nerve fibre density is normal against appropriate genital reference values [8].
- **Claim C refuted.** Body disownership correlates with depersonalisation score and not with the extent or distribution of sensory loss.
- **Claim C weakened further.** Disownership is confined to, or substantially concentrated in, patients with serotonergic antidepressant exposure.
- **Whole conjecture weakened.** Sensory symptoms are present at equal frequency in patients with and without any identifiable nerve injury.
- **Series-wide implication.** If depersonalisation contributes materially, its documented association with *blunted* sympathoexcitation would count against the sympathetic overactivity premise shared across this series.

13. Conclusion

The best thing in this conjecture is a piece of ordinary sensory neurology applied where nobody had applied it. Numbness and dysesthesia in the same territory look contradictory only if one assumes a lesion must be all-or-none. A partial nerve lesion produces both, and that dissolves the tension between a condition whose dominant complaint is sensory loss and a diagnostic spectrum defined by sensory hyperactivity. That insight is worth more than the rest of the conjecture and should be stated on its own.

The rest is weaker than its presentation suggests. The spinal claim adds nothing that two companion conjectures do not already supply with better mechanisms. And the cortical claim — offered, to its author's credit, as the most speculative thing in the source article — rests on the maladaptive plasticity model of deafferentation, which has been under serious challenge since phantom pain was shown to be associated with preserved rather than remapped cortical representation. Building a genital hypothesis on the contested step of a limb literature is not a good foundation.

It is also unnecessary, because a defined condition already accounts for the symptom. Depersonalisation–derealisation involves detachment from one's own body, somatosensory distortion and reduced pain sensitivity; onset is typically between 16 and 23; it is precipitated by stress and distressing illness; a quarter to three quarters of people experience it transiently at some point; and 75% of the largest HFS series had depression or anxiety. There is a validated questionnaire and no HFS study has used it. Separately, 45.5% of that same series were taking anxiolytics or antidepressants, and genital anaesthesia is the cardinal symptom of post-SSRI sexual dysfunction, for which criteria now exist and have never been applied here.

Neither observation means the symptoms are imagined or that patients are unreliable. It means the field is trying to explain a sensory picture without having excluded two common, well-described contributors to exactly that picture — and that excluding them requires a questionnaire and a medication history, not a scanner. Those should come first. Then map the territory, because whether the deficit follows the dorsal nerve of the penis is the one measurement that would inform four of the papers in this series at once.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The depersonalisation and post-SSRI sexual dysfunction literatures were accessed substantially through reference texts and reviews and must be read in the primary sources before these differentials are asserted in print, given their clinical sensitivity.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [6]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Nothing in it should be used to start, stop or change any psychiatric medication;** antidepressant discontinuation without medical supervision carries real risks including relapse and withdrawal phenomena. Raising depersonalisation–derealisation and post-SSRI sexual dysfunction as differential considerations is a statement about what research should measure, not a diagnosis of any individual, and does not imply that reported symptoms are imagined.

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